Big Data in Education: Analyzing Factors That Influence Student Academic Performance

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Abstract

Big Data and learning analytics can enable the exploration of patterns related to students' academic performance and the implementation of timely educational interventions. This paper provides a review of selected research articles published between 2014 and 2025 concerning academic, behavioral, demographic, social, and technology-related student characteristics associated with academic performance and analytical methods utilized to predict it. The original article discusses the presence of prior academic achievement, online learning activity, assessment outcomes, and engagement-related indicators as common elements influencing students' academic performance. The reviewed papers utilize a range of statistical and machine-learning techniques, including logistic regression, support-vector machines, Random Forest, gradient-boosted trees, and deep neural networks, to predict academic performance. However, the performance of these algorithms can vary depending on the available data, definitions, and evaluation metrics utilized, which renders direct comparisons challenging. In this regard, this paper highlights the limitations of algorithm performance reports in the reviewed articles and emphasizes their dependence on data, target definitions, and evaluation approaches. This paper is a review of the relevant literature and does not describe a novel experiment or model training. The study underlines the importance of reporting, model interpretability, privacy, fairness, and the necessity of human interaction in educational analytics. Finally, it offers recommendations related to review procedures, practice, early-warning systems, and interventions.

Index Terms—Big Data, educational data mining, learning analytics, student academic performance, machine learning, early-warning systems, predictive analytics.

I. INTRODUCTION

Educational institutions produce data in the form of information systems, learning management systems, tests, attendance records, and online learning platforms. If harnessed correctly, this data can be utilized to gain insights into the learning engagement of students and those in need of support. Educational data mining (EDM) is concerned with EDM is focused on finding patterns in educational data, whereas learning analytics (LA) is the process of measuring, mining, and analyzing data with the intention of understanding and supporting learning [1], [2].

In terms of predicting academic performance, it is essential to recognize the interplay of several factors and the impossibility of pinpointing a particular set of variables. While past performance and test results of students can predict their academic performance to some extent, engagement, access to resources, motivation, and environment could also affect a student's performance. Predictive models are likely to vary depending on the population of a given institution or data set, and predictions may not always be explainable or may have negative unintended consequences if not deployed with care.

A. Problem Statement

The key problem addressed in this review centers around identifying factors associated with student academic performance as established by the extant literature, and how data-mining and machine-learning methods can be applied to predict such outcomes. One of the issues to be addressed here is how early-warning predictive analytics can produce usable, interpretable, equitable, and privacy-preserving results.

B. Objectives

- Determine academic, behavioral, demographic, social, and environmental factors found to be informative of student performance.
- Summarize and evaluate the methods of analysis utilized in the reviewed works.
- Discuss the implications of learning analytics for the early identification and intervention in the education system.
- Identify methodological, interpretability, privacy, and fairness issues to be considered in the design of future systems.

C. Research Questions

- RQ1: What factors relating to students and the learning context are reported as the most common predictors of academic outcomes?
- RQ2: What statistical and machine learning methods are used for student performance prediction, and what trade-offs are involved?
- RQ3: How can predictions be used for early intervention while ensuring student privacy, fairness, and agency?

D. Scope and Contribution

The review will focus on the research identified in the original project bibliography and described as spanning the years 2014-2025, with certain foundational works included from before this period. The contribution of this review is a synthesis of the factors, methods, applications, and ethical considerations related to educational data mining and learning analytics. As the process of selecting the articles for this review has not been fully documented with specific database statistics in the submitted draft, this manuscript will be framed as a review of the literature rather than a reproducible systematic review.

II. BACKGROUND AND RELATED WORK

A. Educational Big Data and Learning Analytics

Educational big data is formed by structured data objects such as grades and attendance records, semi-structured logs of activities, and traces of interaction in online learning platforms. These data sources enable the creation of complementary perspectives on the learning process, although the value of any given variable depends on its quality, timing, context, and legality of collection. Educational data mining and learning analytics can enable descriptive, diagnostic, predictive, and prescriptive approaches to analyzing and intervening in the learning process [1], [2].

B. Factors Associated with Academic Performance

The surveyed works classify potential predictors of academic performance into loosely defined groups. Prior academic achievement is represented by grades, quiz scores, and assessment results. Attendance, number of visits to the platform, hours of video content viewed, assignments completed, and level of participation are indicators of engagement or student behavior. Socioeconomic status and student demographics are associated with context, while motivation, self-efficacy, social support, access to the internet, and learning resources are linked to process. The presented list is not exhaustive, and each variable should be considered as a potential association rather than a definitive determinant of academic performance.

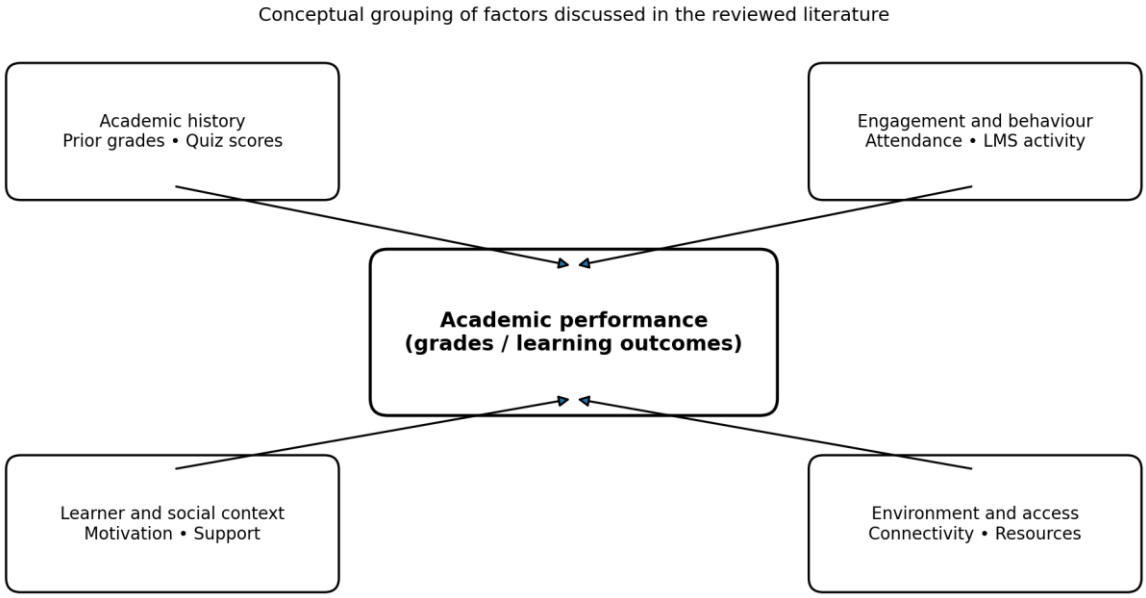


Fig. 1. Conceptual grouping of factors discussed in the reviewed literature.

C. Predictive Modelling Approaches

Common approaches include linear and logistic regression, decision trees, Random Forest, support vector machines (SVM), k-nearest neighbours, gradient-boosted trees, and deep neural networks. Regression models can yield reasonably interpretable relationships, given appropriate assumptions, tree ensembles can represent non-linear interactions, while SVMs can yield good results in high-dimensional settings at the cost of interpretability. Deep-learning models can exploit complex activity sequences, but these may demand substantial data resources and care in interpretation. No single algorithm should be asserted as universally best, without comparable datasets, target definitions, validation procedures, and metrics.

D. Related Research

Namoun and Alshanqiti reviewed 62 studies, finding that online learning activities, assessment grades, and academic emotions were among the evident predictors in the literature [1]. Lu et al. investigated early prediction in a blended Calculus course using variables such as video-viewing behaviour, out-of-class practice, homework, and quiz scores [3]. Waheed et al. analysed virtual learning environment clickstream data, reporting that a deep neural network yielded classification accuracy in the range of 84% – 93% in their experimental setting, depending on the model and data configuration [4]. Ting et al. analysed a large PISA dataset comparing several machine-learning algorithms in student-performance prediction [5]. These results represent the variety of contexts and methods, which should not be directly compared without consideration of the differences in datasets and evaluation designs.

Additional studies in the supplied bibliography present a wider review, beyond the core examples. Khan and Ghosh survey classroom performance analysis and prediction [6], while Kaviyarasi and Balasubramanian discuss potential factors associated with student performance [7]. Kassarnig et al. study academic performance along with behavioural patterns [8], and Yunita et al. combine quantitative and qualitative analyses to identify contributing factors [9]. Cui et al. present a review of predictive-analytic methodologies for student success in higher education [10]. These papers encourage considering academic outcomes in combination with behavioural and contextual information, rather than as a single-factor problem.

The remaining sources in the original bibliography address early-warning approaches and learning behaviour [11], [12], machine-learning analysis of performance factors [13], educational improvement and predictive modelling [14]–[17], and

the analysis of exam results and student performance [18]–[20]. Additional listed work addresses personalized educational data [21], virtual learning environments [22], educational analytics more broadly [23], decision-support systems [24], video-learning analytics [25], multifaceted student-success analysis [26], student retention [27], and general methods for systematic review in educational data mining [28]. These works are retained from the supplied bibliography, however, their full metadata should be verified against the original publisher records, prior to submission.

III. REVIEW METHODOLOGY

A. Review Design

This work implements a literature-review design based on the articles and bibliography cited in the supplied project draft. The existing data was synthesized to produce reported factors, methods, findings, and limitations, instead of collecting a new student dataset or training a new model. The original draft mentions a publication window of 2014-2025 and contains studies reporting on classroom, blended, or online-learning data.

B. Search and Selection Transparency

The supplied project draft does not document the exact dates of the database searches, complete search strings, number of duplicates removed, screening decisions, or the quantity of papers excluded at any stage. This information should be provided before the current work can be classified as a systematic literature review. To enable replication, the search process should be reproduced in scholarly databases such as IEEE Xplore, ACM Digital Library, Scopus, Web of Science, ScienceDirect, and Google Scholar. The process should involve combinations of terms such as “student performance prediction,” “educational data mining,” “learning analytics,” “machine learning,” and “academic achievement.” The final draft should report the databases searched, dates of search, inclusion and exclusion criteria, and the quantities of screened, included, and excluded documents. This section does not contain any unperformed search-count claims.

C. Inclusion and Exclusion Criteria

An appropriate selection protocol would involve including peer-reviewed journal and conference papers focused on student outcomes, reporting educational data or analytical methods, and containing sufficient bibliographic details. The excluded studies would involve documents unrelated to academic outcomes, duplicates, non-scholarly publications, and works with insufficient metadata. Due to the absence of a completed screening log in the supplied file, this set of inclusion and exclusion criteria is suggested for the revised draft, rather than reporting on past screening decisions.

D. Data Extraction and Synthesis

For each included study, the reviewer should report the year of publication, the educational setting or population, the sample or data source used, the definition of outcome variables, the model inputs, the family of models employed, the evaluation procedure, reported effect size or performance metrics, the main findings, and limitations. The findings across similar predictor categories and modelling approaches can then be synthesized thematically, with numerical performance metrics replicated from the original article with evaluation context, rather than pooled across studies.

IV. SYNTHESIS OF FINDINGS

A. Frequently Reported Predictor Categories

Across the literature described in the source draft there are a number of commonly reported predictive features. Online engagement including participation and activity in virtual learning environments can provide insights into the learning process and can potentially enable earlier interventions than those derived from final grades [1], [3], [4]. The relevance of a feature depends on the point in time it is measured; using variables that are recorded after the occurrence of the outcome of interest will lead to information leakage and an overly optimistic estimate of predictive performance.

B. Comparison of Modelling Approaches

Model family	Potential strengths	Important cautions
Logistic / linear regression	Clear baseline; coefficients can be interpretable	Assumptions and feature relationships need checking
Decision trees / Random Forest	Can represent non-linear patterns and interactions	May be harder to explain; calibration and validation matter
Gradient-boosted trees	Often effective on structured tabular data	Tuning and interpretation require care
SVM / k-nearest neighbours	Useful comparison methods for classification	Scaling and parameter choices can strongly affect results
Deep neural networks	Can model complex patterns in large or sequential data	Data needs, transparency, and generalization must be assessed

TABLE I. SUMMARY OF MODEL FAMILIES AND PRACTICAL TRADE-OFFS.

C. Interpreting Reported Performance Metrics

The original draft contains accuracy values from different works and also reports the value of precision. Accuracy is the fraction of correct predictions among all the predictions made; precision is the fraction of predictions of one class that were actually correct; recall is the fraction of actual positive instances that were correctly predicted; F1-score is the harmonic mean of precision and recall. All these metrics answer slightly different questions. In imbalanced problems, such as detecting the educational at-risk, the value of accuracy can be misleading as it summarizes the overall predictive power, while hiding the precision-recall performance on the minority class. When comparing numeric results, it is vital to report the metric definition, target concept, input data, validation approach, data balance, and the class distribution.

For instance, the accuracy reported by Waheed et al. was between 84% and 93% for the considered configuration of their approach, while other works address different data and evaluation settings [4]. Thus, the accuracy values should be treated as evidence for the experiments performed rather than a claim about a generic performance in a different institution.

D. Conceptual Analytics Workflow

An ethical educational analytics workflow assumes that the data collection step was performed under the explicit permission of the data owners and proceeds to the analysis via the validation of the data quality, feature engineering, and model training. The final results should be supported by confidence intervals and proper interpretations for all the stakeholders. The educators should assess the results within the broader context to provide comprehensive support.

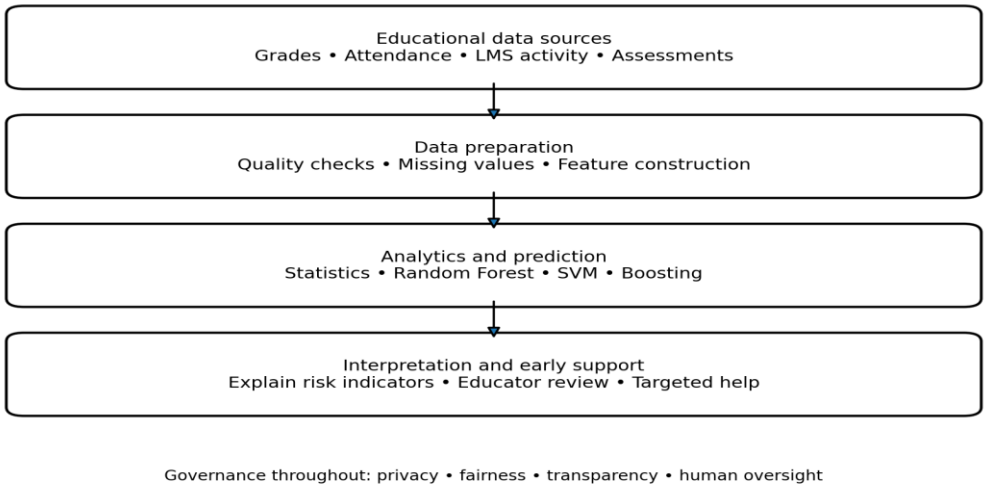


Fig. 2. Conceptual workflow for educational analytics and supportive intervention.

E. Early-Warning Systems

Early-warning systems have the potential to aid institutions by focusing on the timing of conversations, tutoring, and academic interventions. The intention must be aimed at helping rather than punishing. Students should not be permanently tagged by an algorithm and a risk score should never be used as a final word on a student's capabilities or potential.

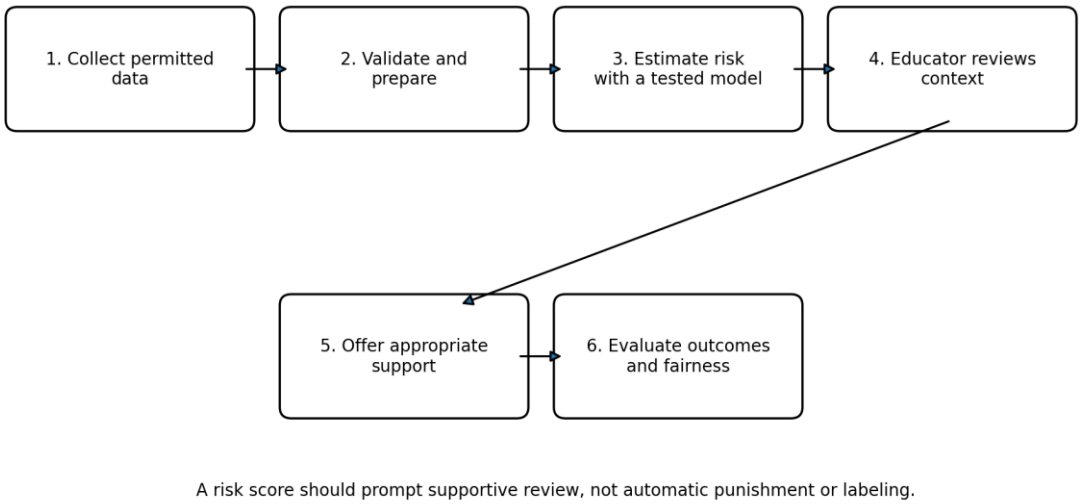


Fig. 3. Responsible early-warning cycle with educator review and outcome evaluation.

V. DISCUSSION

A. Implications for educational institutions

The authors conclude that predictive models based on non-static variables, including prior performance, can be used to create dashboards and early warning processes, as long as such tools are developed and evaluated locally and their shortcomings are acknowledged

The process of creating a predictive model should supplement, not replace, the educational and humanistic elements of teaching, including educator expertise and student discussions.

B. Research gaps

· Temporal and behavioral coverage: some models focus on static features and miss the dynamics of engagement

· Interpretability: teachers need explanations for alerts, including what triggered them, and references for appropriate interventions

· Generalizability: effectiveness depends on context, including institutional characteristics, subject matter, student age, teaching modality, and student diversity

· Fairness and privacy: sensitive attributes, proxies, and disparities in access to technology may be incorporated or inferred, thereby threatening equity and confidentiality

· Evaluation quality: subsequent studies should use non-leakage evaluation methods, report multiple outcome metrics, and assess whether the proposed approach meaningfully improves interventions

C. Ethical and practical considerations

Educational data can include private information about students' behavior, circumstances, and performance. The authors recommend that institutions only collect such data if it is necessary and sufficient for the proposed purpose, secure information through access control and other measures, protect stored data, limit retention periods, and ensure that students understand the process and its implications. Fairness should be evaluated across different groups if such analyses are available, legal, and permissible. The article suggests that a deployed model should continuously be monitored because population shifts, variations in course content or instructors, and changes in the use of competing technologies can affect its performance.

D. Limitations of This Review

This paper was generated using the synthesis already provided in the source document, with additional verification of certain key publications. The first draft did not include a reproducible search log or detailed study-by-study extraction table. As such, this paper cannot claim to have covered all the publications on the topic in 2014-2025, conducted a formal meta-analysis, or verified every single numeric value reported in the studies. The results should be considered a narrative synthesis that requires a documented process of literature search and screening for inclusion in a journal.

VI. RECOMMENDATIONS AND CONCLUSION

A. Recommendations

• Document the search process and the screening, providing the databases, dates, search strings, inclusion rules, and numbers screened.

• Create a verified evidence table for each study and ensure that all performance measures have been reported in their original source.

• Ensure that the comparison is only made between models with a similar target, data context, and evaluation procedure.

• Report multiple metrics, including precision, recall, and F1-score, in the identification of students in need of assistance.

• Emphasize the importance of interpretable data, privacy, fairness, and the necessity of human assessment in the executive summary.

• Evaluate the interventions based on their ability to improve student support and services, not just prediction performance.

B. Conclusion

Big Data and learning analytics can be applied in various ways to gain an insight into the factors that might predict academic performance. As seen from the included articles, prior achievement, assessment data, online activities, and contextual information can be used as input for statistical and machine learning models. All of these predictors have their advantages and drawbacks in terms of performance, simplicity, required data and other aspects. However, the performance of such models cannot be compared directly due to the differences between the studies, and it is essential to remember that the prediction itself does not contribute to the improvement of students' performance. Instead, it is critical to ensure that the intervention done on the basis of the prediction would be beneficial. Thus, the next step for this project is to finalize the inclusion-exclusion process and create a reproducible search record that will be followed by a thorough verification of the listed articles before the submission.

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